

Exploring the Impact of a Liquid-Based System on Full-Body Weight Perception in Weighted Exercise within Virtual Reality

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Abstract

Previous studies have attempted to incorporate weight feedback mechanisms into virtual reality (VR), but most remain limited to localized haptic sensations, with few enabling full-body weight perception. This work explores the impact of a liquid-based wearable system designed to simulate dynamic full-body weight feedback. The system utilizes waterbags and elastic structures to distribute weight across different body regions, enabling up to 10.4 kg of variable load through dynamic liquid transfer. To investigate its effectiveness, we designed an interactive VR experience based on a gym-inspired scenario that simulates weight-shifting sensations during weighted movements. A user study was conducted to evaluate how such a system influences subjective weight perception and immersive engagement in VR.

CCS Concepts

• **Human-centered computing** → **Interaction paradigms**; Virtual Reality.

Keywords

Liquid-based Haptic, Piping System, Weight Sensations, Gravitational Sensations, Virtual Reality

ACM Reference Format:

Chi-Yu Lin, Meng-Wei Lu, Jung Shen, Guan-Yu Zhou, Tse-Yu Pan, and Ping-Hsuan Han. 2025. Exploring the Impact of a Liquid-Based System on Full-Body Weight Perception in Weighted Exercise within Virtual Reality. In

The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25), September 28–October 01, 2025, Busan, Republic of Korea. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746058.3758361>

1 Introduction

To enhance immersion in VR, many studies have attempted to incorporate haptic feedback by simulating the sensation of weight. Most systems adopt a direct weight-adding approach to produce this perception. [3] utilized a water pump to inject liquid into a handheld controller, allowing for a maximum weight of 1 kg with a detachable design, enabling flexible use. [7] and [2] simulated weight changes by adjusting the amount of liquid inside the device, achieving weights of up to 0.9 and 1.5 kg. [1] introduced a wearable device that increases weight using water, with a maximum of 1.8 kg. FlowGlove [5] provided pressure and vibration feedback on the hand using a liquid-filled bladder to simulate haptic sensations. On the other hand, some studies modify body perception or the center of gravity to create a sense of weight. [6] generated accelerated and decelerated jumping sensations by vertically shifting wearable weights, altering the perception of jumping. [4] induced the illusion of arm deformation through skin-stretch feedback, providing a novel form of haptic interaction that is not directly tied to physical weight. This study examines the effect of a liquid-based wearable system on full-body weight perception in virtual reality. By utilizing a reconfigurable dynamic weight-shifting mechanism based on waterbags, the system simulates weight-based haptic feedback across the body. We investigate how this approach influences users' perception and engagement during weighted exercise scenarios in immersive VR environments.

2 DESIGN AND IMPLEMENTATION

In this study, we explored the impact of a liquid-based wearable system on full-body weight perception in virtual reality. The system features a coat-style wearable design, incorporating four 3-liter

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UIST Adjunct '25, Busan, Republic of Korea

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ACM ISBN 979-8-4007-2036-9/25/09

<https://doi.org/10.1145/3746058.3758361>

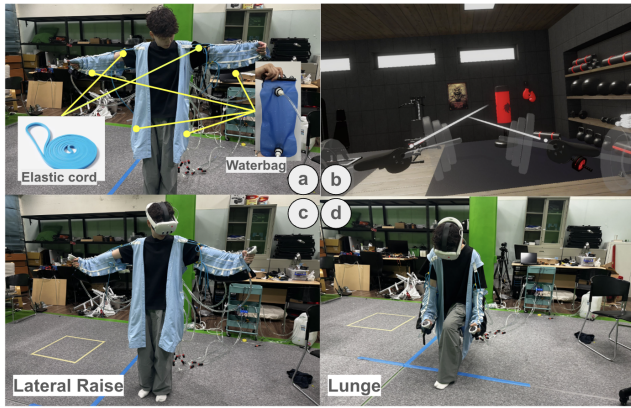


Figure 1: Overview of the Liquid-based wearable System and VR-Based Weighted Training Experiment.(a) Liquid system design, (b) VR gym scene, (c) lateral raise (d) lunge

water bags that can be positioned on the arms and lower body. These bags are connected to the main body using elastic cords that stretch downward as water fills the bags, generating a pulling sensation that supports dynamic weight perception across the body. Controlled by a Linkit 7697 microcontroller, the system receives signals from a computer via Bluetooth to operate water pumps, which move water between a central reservoir and specific water bags. Two independent water loops serve the upper and lower body, providing up to 10.4 kg of additional load. To investigate its experiential effects, we implemented a VR scenario based on a gym environment. The system dynamically adjusted water weight according to interaction phases to simulate realistic weight shifts, enabling an immersive and engaging user experience.

3 User Study

To investigate user interaction with the system, we conducted an exploratory study with 16 participants (8 male, 8 female, aged 18–34). The study combined an immersive VR environment with the dynamic weight feedback by the liquid-based wearable system, aiming to explore how full-body weight perception influences the experience of weighted movements. We selected two test actions: lateral raise, an upper-body movement (Fig.1c), and lunge, a full-body movement (Fig.1d), to observe differences in perceived weight and sense of presence across movement types. For baseline comparison, participants used a pair of 2 kg dumbbells, while the liquid-based wearable system was configured to deliver an equivalent water-based weight. Each participant experienced four interface conditions (VR+Liquid System, NoVR+Liquid System, VR+Dumbbell, NoVR+Dumbbell), performing both movements for 30 seconds under each condition. After each trial, participants rated perceived weight using the Difference Scoring method; for VR conditions, they also completed the Fidelity-based Presence Scale (FPS). A post-study interview was conducted to gather additional feedback.

4 Discussion AND Future Work

This study integrates virtual reality with a wearable liquid-based weight feedback device to explore the impact of full-body weight

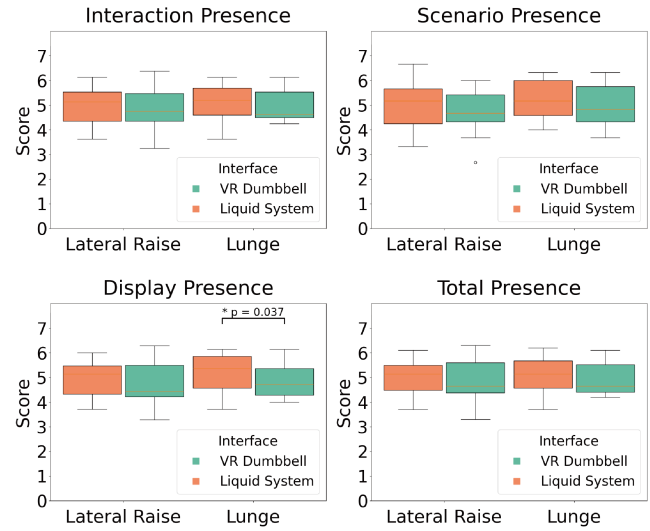


Figure 2: FPS Presence Score Comparison across Interface and Action Conditions

perception on interactive experiences. Through an experiment comparing different interfaces and movement conditions, we analyzed subjective responses. For perceived weight, a Friedman test revealed a significant difference among the four interface conditions during the lunge task (Table 1), whereas no significant difference was found in the lateral raise task. Pairwise comparisons with Bonferroni correction indicated that the VR+Liquid System interface significantly differed from both the NoVR+Dumbbell and VR+Dumbbell conditions (Adj. Sig. = .007 and .013). In the Fidelity-based Presence Scale (FPS) (Figure 2), VR+Liquid System scored slightly higher than VR+Dumbbell across all four dimensions. Further analysis of interface × movement interactions showed no significant differences in most dimensions, except for Display Presence during the lunge task ($p = .037$), suggesting that interface design may affect certain aspects of presence in specific movement contexts. Interview feedback indicated that the Liquid System enhanced the realism of weight sensation and, when combined with VR, strengthened the immersive experience. However, some participants noted that the coat-like form factor might restrict body movement during specific actions. Overall, the Liquid System involves a broader range of body engagement in interactive tasks, allowing users to perceive distributed weight changes across multiple regions, which leads to higher subjective perceptions of weight and presence. When the Liquid-based wearable system provides full-body dynamic weight feedback, the synchronization between visual input and physical sensation is more easily perceived by the whole body, thereby enhancing the sense of presence. This work presents an initial exploration of a full-body interactive system with dynamic weight feedback. Future research will expand to a broader range of interactive movements and conduct more in-depth user studies to improve the system's experiential value and practical applicability.

Table 1: Pairwise Comparisons of Interface Conditions for Subjective Weight Perception (Lunge Action)

Comparison	Test Statistic	Std. Error	Std. Test Statistic	p-value	Adj. p-value
VR+Dumbbell – NoVR+Dumbbell	0.958	5.488	0.175	.861	1.000
VR+Dumbbell – NoVR+Liquid System	-14.083	5.488	-2.566	.010	.062
VR+Dumbbell – VR+Liquid System	-17.792	5.488	-3.242	.001	.007
NoVR+Dumbbell – NoVR+Liquid System	-13.125	5.488	-2.392	.017	.101
NoVR+Dumbbell – VR+Liquid System	-16.833	5.488	-3.068	.002	.013
NoVR+Liquid System – VR+Liquid System	-3.708	5.488	-0.676	.499	1.000

Acknowledgments

This research was partially supported by the National Science and Technology Council of Taiwan under grant 110-2222-E-027-012-MY3 and 114-2425-H-007-002-

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